University College London

PHAS0097: Astrophysics Project

Literature Review

Line List for the N_2 Ground-State Quadrupole Spectrum (ExoMol)

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See Appendix for word count note.

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1 Introduction

The quadrupole spectrum of N_2 provides the only direct spectroscopic signature of molecular nitrogen in *emission*, making it a uniquely valuable atmospheric tracer of bulk composition, despite the absence of a permanent dipole moment. In the absence of high-energy illumination required for UV absorption, these quadrupole transitions remain the only direct means of identifying N_2, yet their extreme weakness has prevented detections beyond laboratory measurements and Earth's atmosphere. With improving observational precision, such features may soon become accessible in planetary and astrophysical settings, offering new constraints on N_2-dominated icy worlds in the Solar System, such as Titan, as well as on terrestrial and super-Earth exoplanets. However, existing line lists are inadequate for such future studies. As part of the ExoMol project, this work aims to produce a comprehensive, temperature-extended line list for all stable isotopologues of N_2, improving atmospheric modelling and helping to distinguish quadrupole features from collisional continuum processes. The report outlines the relevant background, reviews existing datasets, and describes the approach taken to address this gap.

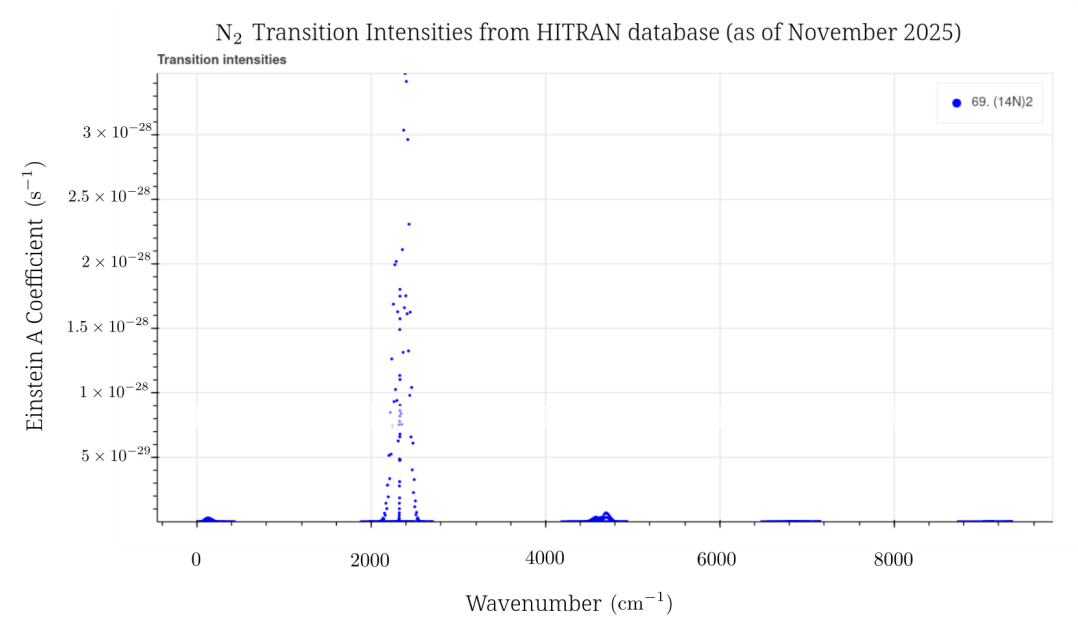

Figure 1: $^{14}N_2$ quadrupole intensities available on the HITRAN database—only the rotational, vibrational, and first three overtone bands are available. Generated by `HAPPI` and visually modified.

2 Literature Survey

2.1 Overview of N_2 spectroscopy

Molecular nitrogen (N_2) is the dominant constituent of Earth's atmosphere, yet its spectroscopy remains challenging and comparatively underexplored. Early investigations focused on its ultraviolet electronic system, which is of limited astrophysical utility. Although UV fluorescence has been observed in the upper atmospheres of Mars [11] and Titan [12], such emission arises from non-thermal processes like aurora, and therefore cannot be used as a reliable tracer of bulk composition. UV absorption, meanwhile, requires a far-UV illuminating source and has yet to be detected in the Solar System or beyond. Moreover, transitions between excited states are substantially more difficult to model than ground-state transitions. By contrast, the rovibrational transitions of the ground electronic state, $N_2(X\,^1\Sigma_g^+)$, offer a tractable route to tracing composition in both emission and absorption, and thus form the focus of this project.

Rovibrational transitions involve simultaneous changes in vibrational (v) and rotational (J) quantum numbers, describing energy levels *approximately* as

$$E_{vJ} = \omega_e(v + \frac{1}{2}) - \omega_e x_e(v + \frac{1}{2})^2 + B_v J(J+1)$$

where ω_e and B_v denote the harmonic and rotational constants. The lack of a permanent dipole moment means the absence of a dipole-allowed infrared spectrum. Instead, N_2 exhibits an electric quadrupole spectrum, governed by $\Delta J = \pm 2$ (rather than ± 1), and is several orders of magnitude weaker, necessitating high-precision instruments and models.

Since the initial laboratory and atmospheric detections in 1954 [13] and 1973 [1], subsequent measurements—culminating in global observations from ACE-FTS [10]—have confirmed the importance of the N_2 quadrupole spectrum in modelling Earth's transmission. In the near term, Solar System targets offer the most realistic prospects for detection: any such measurement must be space-based, as Earth's atmosphere obscures these extremely weak features. Titan's dense nitrogen atmosphere provides a useful analogue for Titan-like exoplanets, while Mars is of interest owing to its elevated $^{14}N^{15}N$ ratio; cometary observations may also constrain nitrogen abundances in primitive bodies. Ultimately, however, the primary scientific goal is to extend such detections beyond the Solar System. With current and upcoming instruments, extra-solar quadrupole signatures may soon be detectable, particularly in dense N_2 atmospheres such as those expected of super-Earths. Accurate interpretation of

these spectra, including separation from the pressure-squared collision-induced absorption (CIA) continuum, will require a high-precision line list.

A line list is a catalogue of all radiative transitions of a molecule, specifying their frequencies and intensities. The HITRAN database [4] is the standard reference, compiling the best available laboratory and theoretical results (shown in Figure 1 for $^{14}N_2$). The current offering, however, is inadequate and will remain unchanged in HITRAN2024, as discussed in the next section.

2.2 Improving line list through ExoMol project

The existing HITRAN2020 database [4] includes only a subset of N_2 quadrupole transitions. For $^{14}N_2$, data extend only to the rotational ($v = 0 \rightarrow 0$), fundamental ($0 \rightarrow 1$), and a few overtone bands up to ($0 \rightarrow 4$) (Figure 1), with even sparser coverage for $^{14}N^{15}N$ (fundamental only) and none for $^{15}N_2$. The more significant limitation is the restricted range of rotational quantum numbers: HITRAN contains quadrupole transitions only up to $J \sim 50$, reflecting its Earth-focused remit. Hot exoplanet atmospheres may populate higher states, and laboratory studies of other stable diatomics show that such levels are accessible; for example, CO has been measured up to $J = 123$ [8]. Since N_2 remains abundant at high temperatures owing to its triple bond, and the $(2J + 1)$ degeneracy favours higher J, its high-J levels may be significantly populated.

This project, as part of ExoMol, aims to generate a complete line list for N_2 extending to dissociation and high rotational levels, applicable across a broad temperature range for all stable isotopes. The dataset will enhance modelling and detection capabilities by retaining even the weakest transitions—an approach justified by phenomena like the unexpectedly amplified radiation-trapped lines in H_3^+ [9]—ensuring future-proof coverage for emerging high-precision instruments. At present, ExoMol includes data only for excited electronic states of $^{14}N_2$, with no coverage of the ground-state quadrupole spectrum.

Two components essentially define a line list: the *potential energy curve* (PEC) and *quadrupole moment function* (QMF). Following review, the available literature data for both are expected to suffice for producing a high-quality line list without further modifications.

The PEC describes how the potential energy varies with internuclear separation, thus defining the transition energies. The most accurate curve available is the direct-potential-fit Morse Long-Range (MLR_4) model by Le Roy [6]. The subscript '4' refers to the fourth-order radial mapping used in the MLR exponent, which gives stable and reliable behaviour for N_2. It reproduces observed term values with spectroscopic

accuracy up to $v = 19$ and extrapolates reliably to high J and towards dissociation. Consistent with ExoMol's principle of "theory when you must, experiment where you can", this potential is derived entirely from experimental data, combining nine independent sources. No *ab initio* calculation yet matches this precision—the most accurate being the r12-MR-ACPF curve of Gdanitz [3]—making the Le Roy MLR potential the preferred choice for this work.

The QMF determines transition intensities through the variation of the quadrupole moment with internuclear distance. The current benchmark *ab initio* function of Li—Le Roy [7] will be adopted here. While direct experimental determination remains difficult, initial cavity ring-down spectroscopy [5] measurements provide strong validation of this QMF.

3 Project Aims

The primary aim of this project is to produce a comprehensive line list for N_2 within the ExoMol framework, extending to dissociation and covering all stable isotopologues. This will form a complete dataset suitable for high-temperature modelling of planetary atmospheres, filling a long-standing gap in both the HITRAN and ExoMol databases. A Gantt chart outlining the project timeline is available in the Appendix.

The line list will be generated using `Duo` [14], ExoMol's nuclear-motion code. `Duo` allows flexible use of analytic potential and moment functions and will be configured here with the MLR_4 potential [6] and the Li–Le Roy quadrupole moment function [7]. The potential will be adapted to `Duo`'s required functional form, while the quadrupole function can be imported directly.

Initial work has focused on familiarisation with `Duo` through example models and documentation. The next step is to validate the potential energy curve by generating ground-state energy levels for several vibrational states and comparing them with HITRAN data. After verification, the nuclear-motion equation will be solved to reproduce existing transitions and then extended to higher v and J levels for all isotopologues.

The resulting line list will be converted into ExoMol format for publication and processed with `pyExoCross` to generate synthetic spectra for verification. Refinement time is reserved to address any discrepancies, including adjustments to the input curves in `Duo` or, if necessary, a limited MARVEL analysis [2].

Stretch goals include incorporating hyperfine structure (dependent on new `Duo` functionality), and exploring whether BOB-induced dipole transitions for $^{14}N^{15}N$ can be included—although these lines are likely to be far weaker than the quadrupole spectrum—before merging the output with existing ExoMol excited-state datasets. These extensions would complete the ExoMol representation of N_2 and enhance its utility for atmospheric and exoplanetary modelling. Time has been allocated for final verification, presentation preparation, and report writing.

4 Conclusion

This document has outlined the motivation for constructing a comprehensive line list for electric-quadrupole transitions of the N_2 ground-state, identified limitations in current HITRAN and ExoMol data, and selected the Le Roy MLR_4 potential and Li–Le Roy quadrupole moment function as suitable foundations. A complete line list will improve modelling of N_2-rich planetary atmospheres by providing accurate quadrupole opacities and a baseline for separating line absorption from CIA continua, ensuring reliable interpretation with current and future high-precision instruments.

So far, `Duo` has been installed and preliminary work has begun on importing and validating these curves. The next stage will generate and verify the full line list, convert it to ExoMol format, and produce synthetic spectra with `pyExoCross`. The accompanying Gantt chart outlines the timeline from setup and validation through refinement and reporting. Supervision meetings will take place weekly or fortnightly, in person.

References

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Appendix

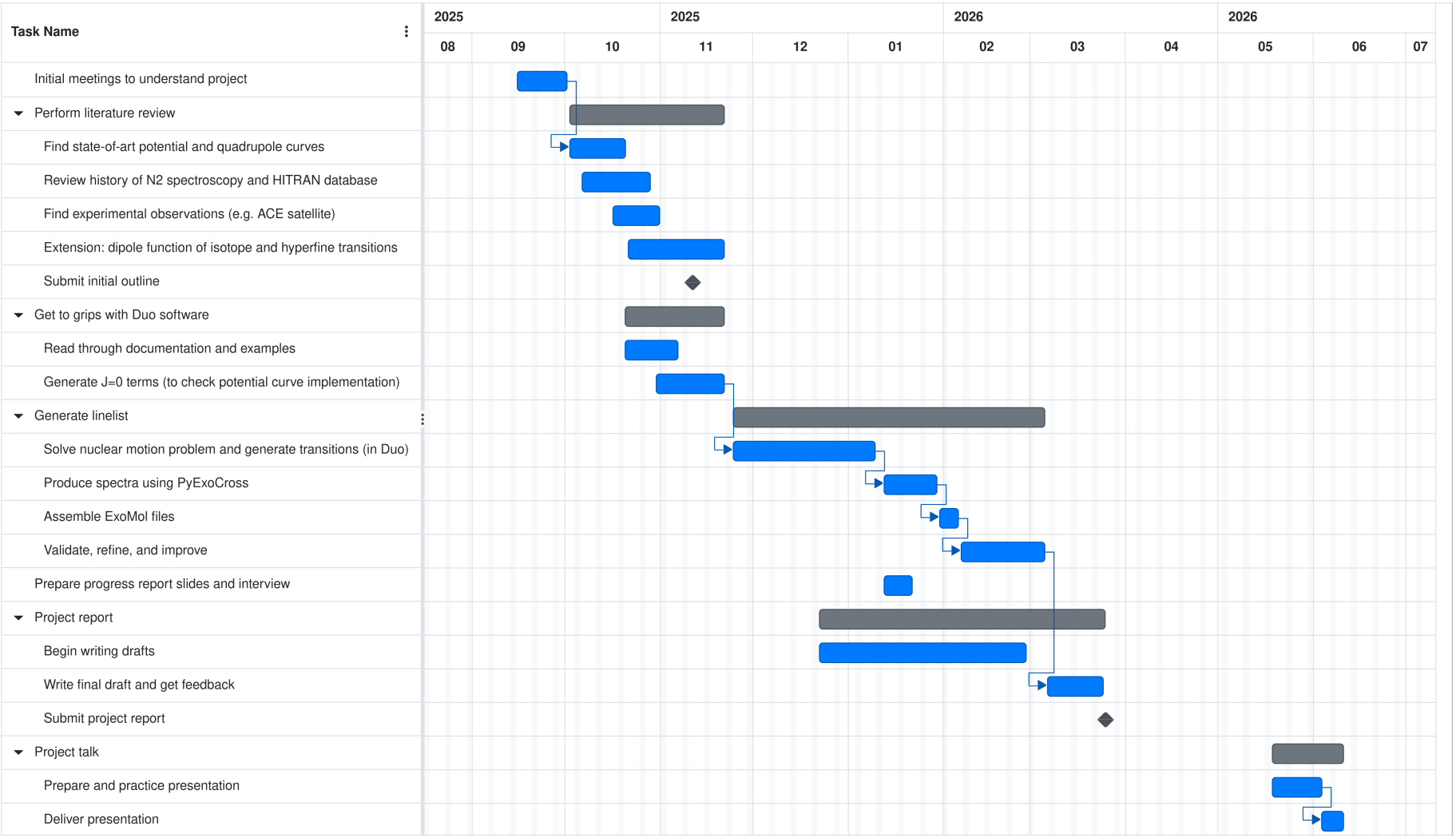

Figure 2: Gantt chart outlining the timeframe for completing main project objectives

Note: I've had a lot of confusion about the word count. I've never had an assignment where the references count towards the word count - this means I've had to include fewer than I normally would. Also, Turnitin seems to count each character in an equation as a word, which doesn't make sense. I propose treating the big equation as two lines (so around 20 words) and treating cases like N_2 as a single word. I propose including heading titles and body in the word count, but not items like the page number, header and cover page. I will speak to my supervisor about the inclusion of references.

Update: my supervisor does not think that the word count that includes everything (including references) leaves enough space to perform a proper literature review, especially since this is a literature heavy project. He proposes excluding the references. He said to include this note.